

Tutorials (SC1006)

Tut 8

Q8.1

(a) Cache block size = $512/32 = 16$ Bytes

Number of main memory blocks = $64\text{KByte}/16\text{Byte} = (2^{16})/(2^4) = 2^{12} = 4096$

(b) Cache Block Size = 16 $\Rightarrow$ Offset Field = 4 bits

Number of blocks in the cache = 32 $\Rightarrow$ Block Field = 5 bits

Main Memory Size = 64KByte $\Rightarrow$ 16 address bits

Tag Field bits = Main Memory Address bits – Offset – Block = 7

$\Rightarrow$ TAG:BLOCK:OFFSET = 7:5:4

Q8.1

(c)

i. $0x0DB63 = 1101\ 1011\ 0110\ 0011$

$\Rightarrow$ Block $10110b =$ Block 22

ii. Number of MM blocks = 4096

Number of cache blocks = 32

$\Rightarrow$ each cache block is a potential destination to $4096/32 = 128$ MM blocks.

iii. By looking at the TAG value entry of the corresponding cache block

For $0xDB63$, the TAG value for the cache block should be equal to

$1101\ 101b$

iv. Offset refers to the offset of the byte of interest from the cache block boundary. For $0xDB63$, the byte is the byte 3 of block 22. First byte of the block is byte 0.

Q8.1

(d)

Assume cache memory and main memory access do not overlap.

$$\begin{aligned} \text{EAT} &= H * \text{Cache}_{\text{Access}} + (1-H) * (\text{Cache}_{\text{Access}} + \text{Mem}_{\text{Access}}) \\ &= 0.9 * 5\text{ns} + (1-0.9) * (5\text{ns} + 200\text{ns}) = 25\text{ns}. \end{aligned}$$

Q8.2

(a) $1\text{Mbyte} = 2^{20} \text{ byte} \Rightarrow 20 \text{ bits}$

(b) $64 \text{ KByte} = 2^6 * 2^{10} = 2^{16} \Rightarrow 16 \text{ bits}$

(c) $\text{Number of virtual pages} = 2^{20} / 2^{10} = 2^{10} = 1024$

$\Rightarrow$ Max number of entries = 1024

(d) $\text{Number of Physical Frames} = 2^{16} / 2^{10} = 64$

$\Rightarrow$ Max number of valid entries = 64

Q8.2 (e)

(i)

0x005F0 → virtual page number 1 → mapped to page frame number 2

0x006FF → virtual page number 1 → mapped to page frame number 2

Virtual Address (20 bits)	Physical Address (16 bits)
0000 0000 0101 1111 0000	0000 1001 1111 0000
0000 0000 0110 1111 FFFF	0000 1010 1111 FFFF

Physical Address = 0x09F0 – 0x0AFF

(ii) 0x009C0 → virtual page number 2 → page fault occurs (valid bit =0).

0x009DF → virtual page number 2 → page fault occurs (valid bit =0).

(iii) Page fault => the required data/code is not in the main memory

=> OS needs to retrieve the required information from the storage memory and update the paging table accordingly.

Q8.2

(f) Main Page Table => Main Memory

TLB => Internal Fast Memory close to CPU

(g) Cache the frequently used Page table information

⇒ Leads to shorter access time for paging information => increase in system performance.

